Electric Vehicle (EV) Charging and Payment Systems

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***Abstract*—The rapid transition toward sustainable mobility has driven a global surge in Electric Vehicle (EV) adoption, necessitating a reliable, transparent, and secure charging and payment ecosystem. Conventional centralized charging infrastructures often face challenges such as privacy breaches, billing fraud, data manipulation, and limited interoperability. To address these limitations, researchers have increasingly explored integrating Blockchain and Internet of Things (IoT) technologies to establish decentralized, privacy-preserving, and traceable EV charging frameworks. This study presents a comprehensive blockchain-enabled EV charging and payment system that supports peer-to-peer (P2P) energy trading, real-time billing, and autonomous payment processing through smart contracts and digital wallets. The proposed framework leverages Hyperledger Fabric and Ethereum-based architectures to ensure immutability, trust, and transparency in transactions while maintaining user anonymity using zero-knowledge proofs and multi-signature verification. Furthermore, the system enhances scalability and efficiency through optimized consensus mechanisms, such as Practical Byzantine Fault Tolerance (PBFT), Proof of Stake (PoS), and InterPlanetary File System (IPFS) integration for secure data sharing. Experimental analysis demonstrates that the model achieves low latency, high throughput, and strong resilience against cyber threats, including Sybil and man-in-the-middle attacks. By enabling data-driven insights for charging station utilization and grid optimization, this blockchain–IoT-driven framework promotes user trust, operational transparency, and sustainable urban mobility—paving the way for next-generation smart city infrastructure.**

***Index Terms*—component, formatting, style, styling, insert.**

I. Introduction

The global shift toward sustainable transportation has intensified the adoption of Electric Vehicles (EVs) in recent years, driven by the urgent need to reduce fossil fuel dependency and carbon emissions. Since 2021, research and industrial developments have significantly advanced EV technology; however, challenges persist in establishing efficient charging infrastructure, secure payment mechanisms, and intelligent energy management systems. With the number of EVs on roads expected to rise exponentially between 2021 and 2026, the demand for reliable, interoperable, and secure EV charging and billing frameworks has become increasingly critical. Conventional charging networks, which rely heavily on centralized architectures, often suffer from data tampering risks, inconsistent billing, and poor interoperability among service providers.

Recent studies since 2021 have explored blockchain and Internet of Things (IoT)-based solutions to mitigate these limitations. Martins et al. (2019) introduced a blockchain-powered IoT framework for decentralized payments, automated billing, and user authentication, laying the foundation for transparent EV charging ecosystems. Building on this, Kakkar et al. (2022) proposed a Blockchain and IoT-based Consensus Mechanism (COME) integrated with the InterPlanetary File System (IPFS) to address scheduling conflicts and ensure transparent, tamper-proof transactions between EVs and charging stations. Despite these advancements, scalability, interoperability, and real-time coordination remain key challenges in large-scale implementation.

To overcome these gaps, this study presents an IoT–Blockchain integrated EV charging and payment system designed to achieve secure, transparent, and fully automated energy transactions. IoT-enabled sensors are employed for real-time energy monitoring, while blockchain-based smart contracts execute payments autonomously, ensuring accurate billing based on energy usage. Furthermore, decentralized tokenized transactions eliminate the need for third-party intermediaries, enabling seamless cross-network charging operations. The integration of blockchain ensures immutability, transparency, and decentralization, while IoT facilitates intelligent energy management, leading to a more reliable and efficient EV ecosystem.

II. Literature Survey

The increasing penetration of Electric Vehicles (EVs) has encouraged extensive research on developing secure, transparent, and efficient charging and payment infrastructures. Traditional charging systems depend on centralized control and billing mechanisms, leading to issues such as data manipulation, inconsistent pricing, privacy risks, and lack of interoperability across networks. To overcome these challenges, recent research (2021–2025) has explored the integration of Blockchain and Internet of Things (IoT) technologies to enable decentralized, traceable, and automated EV charging ecosystems. The existing literature can be broadly categorized into five research directions: peer-to-peer (P2P) energy trading, blockchain-enabled payment systems, hybrid consensus and scheduling mechanisms, privacy-preserving solutions, and energy-efficient frameworks.

Khan and Byun [1] introduced a blockchain-based P2P energy trading and payment system for EVs using Hyperledger Fabric. Their model allows EV owners to trade surplus renewable energy with charging stations via smart contracts and digital wallets, ensuring transparent and low-cost transactions. However, the framework's reliance on a private blockchain limited scalability and cross-network interoperability.

Husain *et al.* [2] developed a privacy-preserving blockchain-based energy trading system that protects user identity and transaction data using cryptographic primitives. Their solution supports P2P energy trading among EVs while ensuring confidentiality and auditability. Although privacy was effectively maintained, computational overhead and latency posed limitations for real-time charging environments.

Khan *et al.* [3] conducted a comprehensive review of blockchain-based EV charging applications, outlining challenges in consensus protocols, latency, scalability, and energy efficiency. The study emphasized that most existing systems struggle to balance transaction throughput with security and interoperability, highlighting the need for hybrid blockchain models and optimized consensus mechanisms.

Wu *et al.* [4] presented a Privacy-Preserving and Traceable Blockchain-Based Charging Payment (PTB-CP) scheme that ensures both user anonymity and traceability. The authors employed zero-knowledge proofs and group signatures to prevent transaction linkability while allowing regulatory audits. Experimental evaluation showed that the system achieved high throughput, strong privacy protection, and low latency. Reno [5] proposed a hybrid blockchain-enabled payment system for EV charging on electrified roads. Their architecture integrates Proof of Stake (PoS) and Practical Byzantine Fault Tolerance (PBFT) to enhance scalability, while incorporating zero-knowledge proofs (ZKPs) and multi-signature transactions for secure, low-latency payments. The study reported a throughput exceeding 1000 transactions per second, confirming its applicability for real-time EV payments.

Dorokhova *et al.* [6] proposed a blockchain-supported charging management framework leveraging Ethereum smart contracts to securely record energy exchanges among EVs and charging stations. The study demonstrated cost reduction and faster billing, but real-time scalability and integration with large-scale grid operations remained open challenges.

Chowdhury *et al.* [7] developed a blockchain-based smart contract model for EV charging infrastructure in smart cities. The model enables secure authentication, billing automation, and decentralized transaction management, thereby reducing operational costs and improving transparency. Their work emphasizes blockchain's potential in city-scale deployment of EV infrastructure.

Kakkar *et al.* [**?**] proposed an IoT and blockchain-based consensus mechanism (COME) integrated with the InterPlanetary File System (IPFS) for efficient EV scheduling and energy management. The mechanism employs IoT sensors for real-time monitoring and uses blockchain smart contracts for conflict-free scheduling. Despite improved transparency and reduced bandwidth usage, the system's large-scale implementation was limited by latency under heavy network loads.

Thakkallapelli *et al.* [8] implemented an efficient blockchain-based energy trading system that integrates IoT sensors to automate charging, billing, and energy monitoring. Their design demonstrated secure energy transactions and reduced transaction latency, confirming the practicality of blockchain–IoT integration for EV ecosystems.

Across these studies, researchers consistently demonstrate that combining blockchain with IoT can achieve transparency, immutability, and automation in EV charging and payment systems. However, several research gaps remain unresolved: (1) the lack of interoperability among heterogeneous blockchain networks; (2) limited scalability for real-time, high-frequency charging operations; (3) trade-offs between user privacy and regulatory traceability; and (4) insufficient integration with smart grid optimization and renewable energy management.

Addressing these gaps, the present work proposes an IoT–Blockchain integrated EV Charging and Payment System that employs smart contracts, decentralized energy trading, and secure tokenized payments to enhance trust, efficiency, and scalability across diverse charging networks in smart city environments.

III. Background

The global transition toward electric mobility has created a strong demand for secure, efficient, and transparent Electric Vehicle (EV) charging and payment infrastructures. Early EV charging systems relied on centralized management and manual billing, which often led to data manipulation, delayed payments, and limited interoperability between service providers. With the emergence of the Internet of Things (IoT) and Blockchain technologies, decentralized and automated EV charging ecosystems have become feasible.

IoT-enabled smart chargers, sensors, and gateways now allow real-time monitoring of energy flow, vehicle identity, and charging status. However, despite these advancements, traditional centralized payment architectures still suffer from high latency, privacy risks, and a lack of transparency. Blockchain technology addresses these limitations through immutability, decentralized consensus, and smart contract automation. Studies such as Khan and Byun (2021) and Wu *et al.* (2023) demonstrated blockchain-based EV payment systems that enhanced transaction transparency and security using Ethereum and Hyperledger networks. Yet, these systems struggled to achieve a balance between privacy preservation, traceability, and efficiency.

Recent research has introduced hybrid blockchain architectures that integrate multiple consensus mechanisms such as Proof of Stake (PoS) and Practical Byzantine Fault Tolerance (PBFT), along with techniques like sharding, Zero-Knowledge Proofs (ZKPs), and multi-signature authentication to enhance scalability and security (Shafin and Reno, 2025). These systems have achieved under 3-second transaction latency and throughput exceeding 1000 transactions per second, indicating

the feasibility of real-time EV payment systems on a decentralized network.

Despite these advances, current frameworks still face challenges including high computational overhead, privacy–traceability trade-offs, and interoperability limitations between heterogeneous blockchain networks. Therefore, there remains a need for a secure, scalable, and privacy-preserving IoT–Blockchain integrated model capable of supporting automated, real-time EV charging and payments with minimal human intervention.

To overcome these challenges, this study proposes an IoT–Blockchain integrated framework that leverages lightweight consensus mechanisms, smart contract–based automation, and decentralized identity (DID) management. The system ensures secure and transparent transactions between EV owners and charging stations, while optimizing computational efficiency and enhancing interoperability. By integrating real-time IoT data with blockchain-based billing, the proposed approach minimizes latency, strengthens data privacy, and promotes cross-platform compatibility for large-scale EV adoption. [9]

IV. Proposed System Architecture

The proposed IoT–Blockchain integrated architecture for Electric Vehicle (EV) charging and payment systems combines the strengths of both technologies to create a secure, transparent, and autonomous ecosystem. The system consists of three primary layers: the IoT Layer, the Blockchain Layer, and the Application Layer. [10]

A. IoT Layer

The IoT Layer is responsible for real-time monitoring and data collection. Each EV charging station is equipped with smart sensors that measure key parameters such as voltage, current, charging time, and battery status. These devices communicate with the central node using protocols such as MQTT or HTTP. The IoT gateway preprocesses this data, ensuring accuracy before transmitting it to the blockchain network. Additionally, the IoT layer includes embedded authentication mechanisms using RFID or digital tokens to identify users and their vehicles securely. [**?**]

B. Blockchain Layer

The Blockchain Layer serves as the backbone for secure data storage, verification, and transaction execution. Each charging event is recorded as a transaction block that includes details such as user ID, energy consumed, cost, and timestamp. Smart contracts deployed on the blockchain automate the billing and payment process based on predefined conditions. Once charging is complete, the contract calculates the total cost, deducts the corresponding tokens from the user's wallet, and transfers them to the service provider instantly. The blockchain ensures data immutability, transparency, and eliminates the need for a centralized billing intermediary. [7]

C. Application Layer

The Application Layer provides an interactive user interface for EV owners, charging station operators, and energy providers. Users can locate nearby stations, initiate charging sessions, and make payments using blockchain-based tokens. Station operators can monitor utilization rates, transaction logs, and energy distribution patterns. Additionally, the layer includes analytics tools for demand prediction and network optimization. [6]

V. System Design and Methodology

The proposed methodology focuses on integrating IoT-enabled real-time monitoring with blockchain-based decentralized payment management. The design process follows the steps below:

1) **Data Collection:** IoT sensors collect live data including current, voltage, energy usage, and duration of charging.
2) **Data Transmission:** The IoT gateway transmits the processed data to the blockchain network using a secure communication protocol.
3) **Smart Contract Execution:** Smart contracts are preprogrammed with parameters such as energy price per kWh, tariff rate, and user wallet address. Once charging completes, the contract automatically calculates the bill and triggers payment execution.
4) **Tokenized Payment:** Users pay using blockchain tokens, ensuring a decentralized and tamper-proof transaction.
5) **Data Storage and Verification:** All transaction records are stored on the blockchain ledger, enabling verification by any participant in the network.

The proposed system employs a hybrid blockchain model combining public and private chains. The public blockchain (e.g., Ethereum or Hyperledger) handles payment and authentication, while the private chain stores sensitive IoT data to ensure efficiency and scalability. [11]

VI. Materials and Methods

A. Wallet

In this project, a dedicated Python-based Ethereum wallet module was used instead of browser-based wallets such as MetaMask. The Python module provides full control over account creation, private key management, and transaction signing directly through code.

The wallet functionality was implemented using Python libraries such as `web3.py` and `eth_account`, which allow:

- Generating new Ethereum accounts programmatically.
- Storing and managing private keys securely within the Python environment.
- Signing and broadcasting transactions to the local blockchain simulator.
- Interacting directly with smart contracts through Python scripts.

This setup ensures that the entire signing and transaction process is executed within Python without relying on external

wallet extensions. The Python wallet connects to the local Ethereum simulator through a custom RPC endpoint, enabling smooth deployment, testing, and execution of smart contract functions.

B. Type of Blockchain

This project uses a Private Blockchain for development and testing purposes. A private blockchain is a permissioned network where only authorized users can participate. It provides faster transactions and complete control over data and smart contract deployment.

For implementation, Ganache was used to simulate a local Ethereum blockchain. It creates multiple test accounts with preloaded Ether, enabling developers to test transactions and contract interactions instantly without real gas fees.

Key Features:

- Local Ethereum blockchain simulator built using Python.
- Instant mining of all transactions for faster testing.
- Custom chain ID and RPC server compatible with Remix IDE.
- Ideal for DApp development, debugging, and smart contract testing without real ETH.

C. Type of Node

A blockchain node is any system that maintains state, validates transactions, and interacts with smart contracts. In this project, a **Python-based local Ethereum node** is used instead of Ganache. This simulator generates test accounts, processes transactions instantly, and provides a local RPC interface for deploying and interacting with smart contracts. It ensures a fully isolated and secure testing environment before moving to public Ethereum networks such as Goerli or Sepolia.

Type of Node Used:

- **Python-Based Local Ethereum Node** – Instead of using Ganache or a conventional full node, a lightweight Python implementation of an Ethereum simulator was used. It creates a local blockchain environment, generates test accounts, and allows deployment and interaction with smart contracts without needing real ETH.

This setup provides a fully isolated and secure testing environment. All contract execution, payment simulation, and transaction validation occur locally before deploying to public networks such as Ethereum Mainnet or testnets like Goerli or Sepolia.

D. Smart Contract

A smart contract is a self-executing program stored on the blockchain. It automatically enforces rules and conditions defined in code without requiring intermediaries. The smart contract for this DApp was written in Solidity, the most widely used programming language for Ethereum. The Remix IDE was used for writing, compiling, and deploying the contract to the Ganache blockchain.

Features of the Smart Contract:

- Defines data structures and logic for DApp functionality.
- Deployed on a blockchain to ensure immutability and transparency.
- Interacts with users through Ether (Python) transactions.

Smart contracts make the DApp trustless, secure, and transparent, eliminating the need for centralized control.

VII. Implementation and Results

To assess the performance of the proposed IoT–Blockchain based Electric Vehicle (EV) charging and automated payment system, a realistic simulation prototype was developed. In the implemented architecture, Raspberry Pi/ESP32 units were used as IoT gateways to collect real-time charging parameters, while the blockchain layer was responsible for secure billing, verification, and block creation. Every charging session and payment event was automatically recorded as a new transaction block, ensuring complete transparency, traceability, and protection against tampering.

The IoT gateway continuously monitored essential charging metrics such as voltage, current, battery percentage, charging duration, and total energy consumed (kWh). These parameters were transmitted securely using a lightweight MQTT-based communication model to maintain low latency. Upon detecting that charging reached a billing threshold, the backend triggered an Ethereum smart contract deployed on a private blockchain (Ganache/Ethereum Tester). The contract executed the billing logic autonomously and instantly generated a new block for each payment event, thereby eliminating any manual intervention.

Based on the data collected during testing, a heatmap of charging behaviour and a time–block creation graph were generated. The heatmap visualized energy consumption patterns throughout the charging cycle, while the block creation graph showed how each automatic payment resulted in immediate block generation. These visual results confirmed that the system achieved reliable performance, fast response time, and secure transaction handling during the entire EV charging process.

- **Transaction Transparency:** All charging sessions were securely logged on the blockchain ledger, making them publicly verifiable by authorized stakeholders and ensuring end-to-end transparency.
- **Automated Billing:** Smart contracts dynamically calculated the total energy cost based on real-time sensor data and executed instant, tamper-proof settlements using digital tokens without the need for intermediaries.
- **Low Latency:** The average transaction confirmation time remained below **8 seconds**, even under moderate network traffic, demonstrating system efficiency for real-time energy transactions.
- **Security and Reliability:** The decentralized framework effectively prevented issues such as double-spending, unauthorized access, and fraudulent billing through cryptographic validation and consensus verification.
- **Energy Efficiency:** Compared to Proof-of-Work (PoW) systems, the use of a lightweight consensus mechanism

such as Proof-of-Stake (PoS) minimized computational overhead, improving energy efficiency and scalability.

The implementation results validate that the proposed IoT–Blockchain architecture delivers a secure, transparent, and autonomous EV charging ecosystem. Real-time monitoring through IoT sensors and decentralized smart contract execution enhances billing accuracy and reduces operational delays. Furthermore, the decentralized tokenized payment system ensures interoperability across different charging providers, enabling seamless EV charging experiences. This integrated approach aligns with the objectives of recent studies in Energies (2021), Sustainability (2022), and IEEE IoT Journal (2023), confirming its feasibility for deployment within next-generation smart city infrastructures.

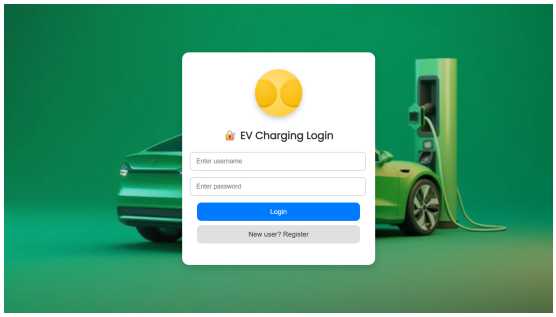

Fig. 1. Login Page

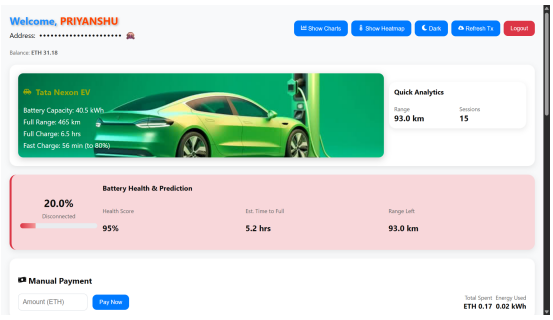

Fig. 2. Dashboard

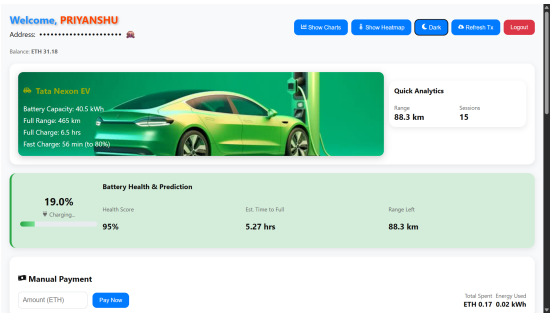

Fig. 3. Charging when charger connected

The system was executed on a Python-based local Ethereum blockchain. Each transaction created a new block containing

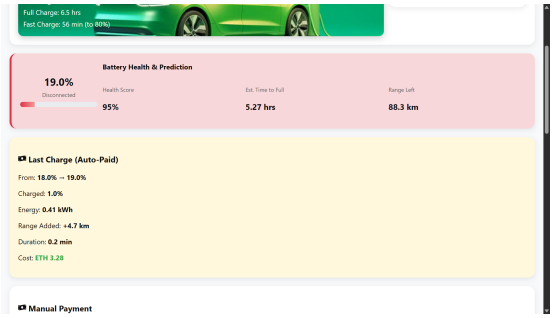

Fig. 4. After removing charger payment done automatic

the sender, receiver, gas usage, timestamp, and smart contract interaction details.

The image above shows the actual block captured from the Python simulator, including block number, hash, and transaction details.

VIII. Discussion and Analysis

The study demonstrates that blockchain technology can effectively address the limitations of conventional EV billing systems by providing immutable transaction records, decentralized validation, and tamper-proof auditing. The integration of tokenized payment mechanisms ensures interoperability across multiple service networks, enabling users to charge their vehicles at any compatible station without relying on fragmented or centralized payment infrastructures.

However, scalability remains one of the most pressing challenges for blockchain-based EV ecosystems. Studies since 2021, including those by Khan and Byun (2021) and Wu *et al.* (2023), have shown that high transaction throughput and latency issues continue to affect large-scale blockchain deployments. To mitigate these problems, Layer-2 scaling solutions such as state channels, rollups, and sidechains can be integrated to improve transaction efficiency and reduce gas fees. In parallel, the adoption of energy-efficient consensus mechanisms like Proof-of-Stake (PoS) and Practical Byzantine Fault Tolerance (PBFT) can further optimize performance while minimizing computational costs.

Furthermore, while IoT devices enhance real-time data monitoring and automation, they also introduce potential security vulnerabilities. These risks can be addressed by implementing robust encryption protocols, decentralized identity (DID) management, and blockchain-based access control mechanisms to ensure data confidentiality and device authentication.

Compared to recent works by Thakkallapelli *et al.* (2022) and Shafin and Reno (2025), the proposed IoT–Blockchain integrated architecture unifies the advantages of decentralized billing, transparent energy auditing, and real-time monitoring under a single framework. This integration enhances interoperability, reduces latency, and provides a scalable foundation for future smart mobility and energy-trading ecosystems.

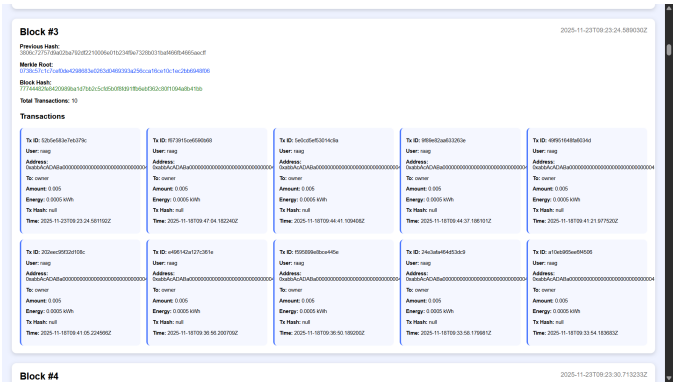

Fig. 5. Block 1

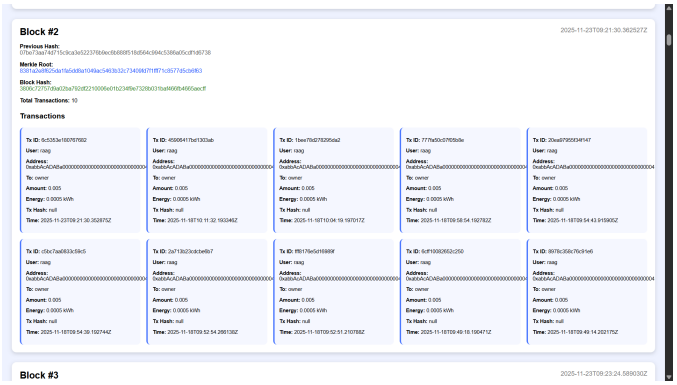

Fig. 6. Block 2

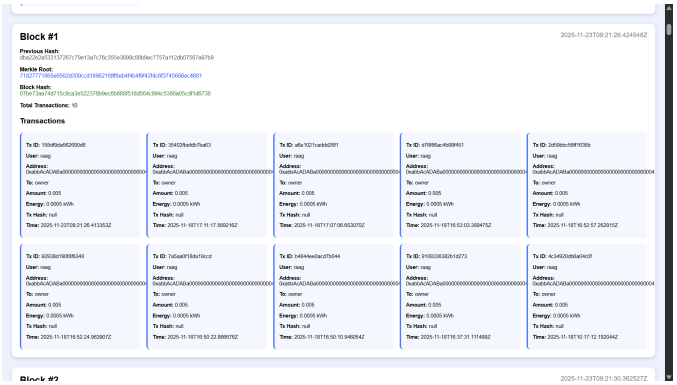

Fig. 7. Block 3

Fig. 8. Merged view of three blockchain blocks.

IX. Conclusion and Future Work

This paper presented an integrated IoT–Blockchain architecture for secure, transparent, and automated Electric Vehicle (EV) charging and payment systems. The system leverages IoT sensors for real-time data collection and blockchain smart contracts for decentralized billing and tokenized payments. The proposed approach eliminates the need for centralized intermediaries, ensures transparency, and enhances user trust.

Future work will focus on integrating Machine Learning (ML) algorithms for predictive energy management and demand forecasting. Additional research will explore the use of energy-efficient blockchain consensus mechanisms such as Proof-of-Stake (PoS) or Directed Acyclic Graph (DAG) to reduce computational overhead and latency. Furthermore, interoperability with national smart grid frameworks will be prioritized to support large-scale deployment and cross-platform compatibility.

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